

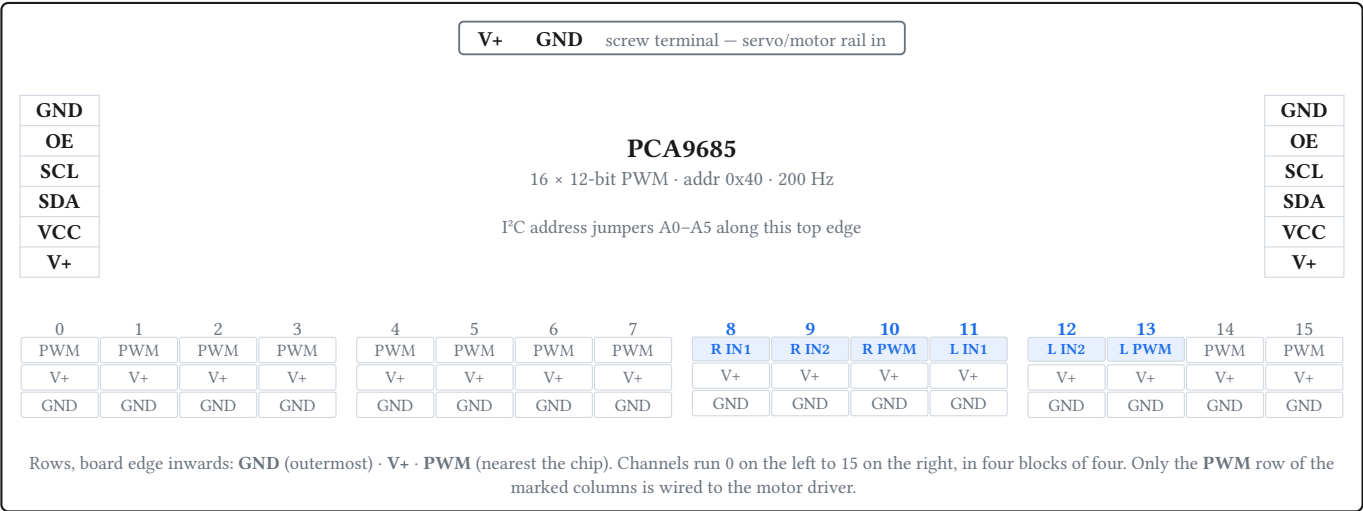
Motor wiring card PCA9685 → TB6612FNG · robocar-unified

Six signal wires, plus one that does not come from the PCA9685

Channels 8–13 carry direction and speed for both motors. **STBY** is the seventh wire and runs straight from the XIAO’s **D0 / GPIO1** — it is not a PCA9685 channel. With STBY low the H-bridges stay disabled and the motors do nothing, however correct the other six wires are.

1 · Board layouts, viewed from the component side

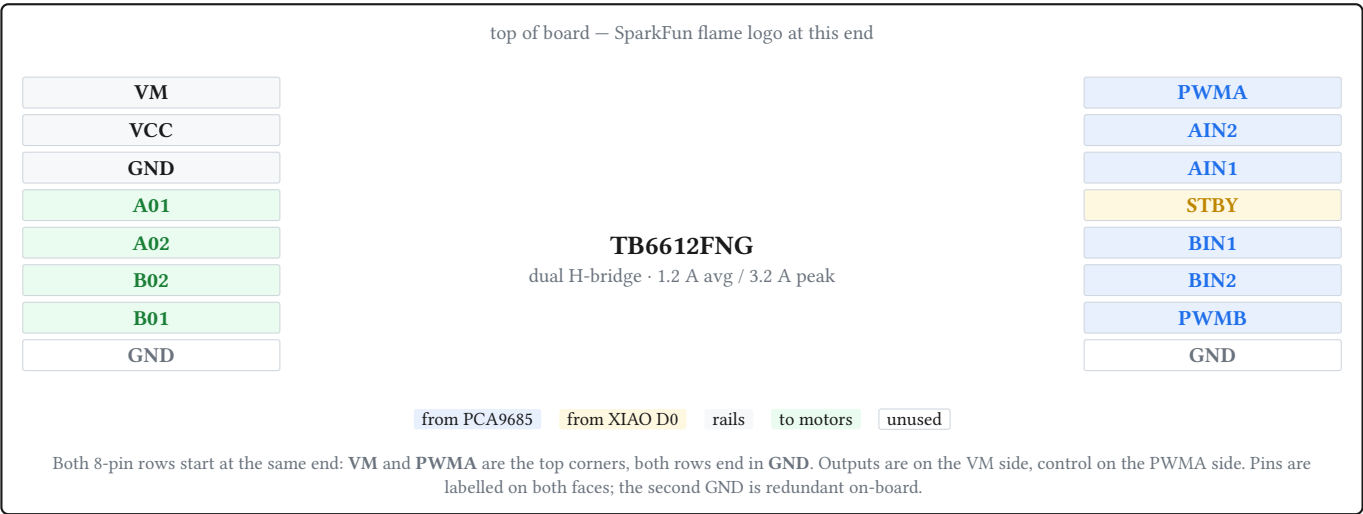
PCA9685 16-channel PWM driver — Adafruit #815 and its clones



Both side headers carry the same six pins

The 6-pin headers on the left and right edges are identical — GND, OE, SCL, SDA, VCC, V+ top to bottom — so either one can take the I²C feed and the other chains onward. Here the I²C side comes from **TCA9548A channel 0**, not from the XIAO directly. The board’s own underside warns: **the screw terminal is reverse-polarity protected, the side breakout pins are not** — the protection MOSFET sits behind the terminal only, so V+ fed backwards through a side header kills the board. **OE** needs nothing: a 10 kΩ pulldown on the board already enables the outputs.

TB6612FNG dual motor driver — SparkFun ROB-14451 and its clones



A01 / A02 / B01 / B02 are printed with a zero, not a letter O

Channel **A** drives the right motor and **B** the left, matching docs/schematics/circuits/robocar_unified.py and the MOTOR_RIGHT_* / MOTOR_LEFT_* names in main/pin_config.h. Nothing in the firmware can tell which motor is which — that is set entirely by which output pair you solder to which wheel.

2 · The seven wires

	From	To	Carries
☐	PCA9685 ch 8 · PWM row	TB6612 AIN1	Right motor direction bit 1 — driven full-on (4096) or full-off (0), never a duty cycle
☐	PCA9685 ch 9 · PWM row	TB6612 AIN2	Right motor direction bit 2
☐	PCA9685 ch 10 · PWM row	TB6612 PWMA	Right motor speed — the only modulated line of the three
☐	PCA9685 ch 11 · PWM row	TB6612 BIN1	Left motor direction bit 1
☐	PCA9685 ch 12 · PWM row	TB6612 BIN2	Left motor direction bit 2
☐	PCA9685 ch 13 · PWM row	TB6612 PWMB	Left motor speed
☐	XIAO D0 / GPIO1	TB6612 STBY	Global enable, straight from the MCU — high enables both bridges

Leave the V+ and GND rows of those six columns unconnected

Each PCA9685 channel breaks out three pins, but the motor driver takes only the signal. The V+ and GND pins of columns 8–13 stay empty; the driver gets its power and its ground from the rails below, not from the PWM header.

3 · Power and ground

	Pin	Rail	Note
☐	TB6612 VM	5.0 V from the LM2596	Motor supply. Star-wire it from the regulator's output terminal — do not daisy-chain off the amplifier or the servo feed
☐	TB6612 VCC	3.3 V	Logic supply. Must match the PCA9685's VCC — see §4
☐	PCA9685 VCC	3.3 V	Logic supply, via the side header
☐	PCA9685 V+ (screw terminal)	5.0 V from the LM2596	Servo rail. 6 V absolute maximum, per the board's own silkscreen. Fit $\geq 470 \mu\text{F}$ here
☐	TB6612 GND (either pin)	Common ground	Both GND pins are joined on-board; one wire is enough
☐	PCA9685 GND	Common ground	Terminal-block GND and header GND are the same net

A missing ground does not stop a chip from running

A board whose GND is not connected finds a return path through the ESD clamp diodes on its signal pins and keeps working — badly, in a way that reads as a firmware bug. Buzz every ground to the regulator's negative terminal **before** applying power, not after something misbehaves.

4 · Why both logic rails are 3.3 V

Both parts specify their input threshold as a fraction of **their own** supply, so the rail is not a free choice:

Part	Spec	Consequence
PCA9685 SCL/SDA	$V_{IH} = 0.7 \times V_{DD}$	At $V_{DD} = 5 \text{ V}$ the threshold is 3.5 V, above what the XIAO's 3.3 V I ² C can drive. At 3.3 V it is 2.31 V
TB6612 IN1/IN2/PWM	$V_{IH} = 0.7 \times V_{CC}$	At $V_{CC} = 5 \text{ V}$ the threshold is 3.5 V, above the 3.3 V the PCA9685 would output
TB6612 STBY	$V_{IH} = 0.7 \times V_{CC}$	At $V_{CC} = 5 \text{ V}$ a 3.3 V GPIO cannot reliably lift STBY — the motors would stay in standby

WIRING.md currently says 5 V here — the schematic says 3.3 V

WIRING.md's power diagram feeds 5 V to both VCC pins; the schematic at docs/schematics/circuits/robocar_unified.py feeds 3.3 V. The arithmetic above settles it in favour of 3.3 V for both, with the 5 V rail reaching only TB6612 VM and PCA9685 V+. Wire it that way and treat the diagram as the stale copy.

5 · Before power-up, and first motion

Check	Expected
☐ Continuity, every GND pin to the regulator's negative terminal	Beeps — all of them
☐ Continuity, TB6612 VM to TB6612 VCC	No beep. A beep means the 5 V and 3.3 V rails are shorted
☐ Regulator output, unloaded, before anything is connected	5.0 V — the LM2596 trimpot spans 1.2–37 V and arrives set at neither
☐ servo on the console	Reports the live PWM frequency — proves the PCA9685 answers on TCA9548A ch 0 before any motor moves
☐ F on the console, wheels off the ground	Both wheels forward for about a second, then stop
☐ trace on the console	I ² C op counter climbing while driving, near-idle when stopped

Truth table, per channel

IN1	IN2	PWM	Mode
H	L	H	CW
L	H	H	CCW
H	L	L	Short brake
H	H	—	Short brake
L	L	H	Stop (coast)

STBY must be high for any of it. The firmware drives forward as IN1 = high, IN2 = low.

If a wheel runs backwards

Swap that motor's **two output wires** at A01/A02 or B01/B02. Do not swap the IN1/IN2 wires and do not change the firmware: the direction bits are shared with the reactive controller's turn logic, so inverting them there fixes driving straight and breaks turning.

If nothing moves at all

Measure STBY. Below 2.31 V with V_{CC} at 3.3 V, the bridges are held off and every other wire is irrelevant.

Sources. Channel numbers and the STBY GPIO: `main/pin_config.h` via `docs/auto/pin_defs.typ`, regenerated and diffed by `.github/workflows/build-guide-check.yml`. Board layouts are read out of the vendors' published Eagle files — Adafruit `Adafruit-16-Channel-PWM-Servo-Driver-PCB` (Adafruit PCA9685 rev C.brd) and SparkFun `Motor_Driver-Dual_TB6612FNG` (SparkFun_Motor_Driver-TB6612FNG_v11.brd) — not from a product photo. Thresholds: NXP PCA9685 datasheet rev. 4 and Toshiba TB6612FNG datasheet, both listed in `docs/reference/datasheets/`. Rails and decoupling: `WIRING.md` § Power.